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Heterogeneous nitration of nitrobenzene in microreactors: Process optimization and modelling

Nan Jin [a,*], Yibing Song [a], Jun Yue [b], Qingqiang Wang [a], Pengcheng Lu [a], Yaoyao Li [a], Yuchao Zhao [a,*]

[a] Shandong Provincial Key Laboratory of Chemical Engineering and Process, College of Chemistry and Chemical Engineering, Yantai University, Yantai 264005, China
[b] Department of Chemical Engineering, Engineering and Technology Institute Groningen, University of Groningen, Nijenborgh 4, 9747 AG Groningen, The Netherlands

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ABSTRACT

Aromatic nitration with mixed acid is an important step in the industrial synthesis of basic industrial chemicals, while the highly exothermic and heterogeneous nature of the reaction renders difficulties in achieving high safety and efficiency. In this work, a continuous flow microreactor system was developed for the nitration of nitrobenzene. Under the premise of high conversion and selectivity, the reaction time and temperature were reduced from more than 2 h and 80 °C in industrial operation to 10 min and 65 °C in microreactors, respectively. The reaction mechanism was verified that the reaction took place not only in the bulk aqueous phase but also at the phase interface. The kinetic parameters were then determined based on the assumption of pseudo-homogeneous reaction mixture. These findings may shed important insights into the high degree of controllability of nitration for a better process design and reactor optimization.

1. Introduction

Aromatic nitration is an important reaction in the production of basic and specialty chemicals, with applications in fields such as dyes, pharmaceuticals, and agrochemicals, etc. (Patel et al., 2021) In majority of these reactions, aromatic hydrocarbon is nitrified into nitroaromatic compound by mixed acid (the mixture of sulfuric acid and nitric acid), in which nitric acid acts as a nitrating agent and sulfuric acid as a catalyst. Reports of experimental studies on nitration of aromatics (such as benzene, toluene, chlorobenzene, benzaldehyde etc.) by mixed acid are available in the literature (Burns and Ramshaw, 2002; Liang et al., 2022; Quadros et al., 2004, 2005). It is commonly known that nitration is a fast and highly exothermic reaction, with the reaction heat ranging from 73 to 253 kJ/mol (Kulkarni, 2014). Therefore, it is essential for nitration reactors to provide adequate heat removal rate, otherwise uncontrollable side or runaway reactions can occur, leading to safety hazards (van Woezik and Westerterp, 2002; Zhou et al., 2014). In batch or continuous stirred-tank reactors extensively used in nitration processes, optimizing the design of stirring paddles and cooling coils to enhance the rate of heat removal, as well as employing relatively low reaction temperature or adding the mixed acid in batches to slow down the reaction rate (and with that the heat generation), are the most common methods to reduce safety risks (Gao et al., 2020; Song et al., 2022c; Xu et al., 2018; Zhang et al., 2015). However, this is usually at the cost of long reaction time (typically tens of minutes or even several hours), which would result in not only the reduction of reaction efficiency, but also the increase of by-product formation (Gao et al., 2020; Xu et al., 2018). Therefore, it is of high importance to explore novel reactors for aromatic nitration that can boost the reaction efficiency with much improved process safety.

Recently, microreactor technology has opened new horizons both in academia and industries, becoming a more sustainable and reliable alternative for safer nitration (Cui et al., 2022; Ducry and Roberge, 2005; Guo et al., 2023; Li et al., 2022; Song et al., 2022a; Song et al., 2022b). Microreactors possess internal channels with micron or sub-millimeter characteristic dimensions, based on which many advantages have emerged, ranging from excellent mass and heat transfer performance to easy scale-up and potential of full process automation. Ducry et al. (Ducry and Roberge, 2005) found that the temperature rise during the nitration of phenol in the microreactor was measured to be no more than 5 °C, which was much lower than that in a semibatch reactor (i.e., 55 °C). The efficient heat transfer capability of the microreactor makes it possible for the elimination of local "hot spots", allowing safe control of the exothermic nitration. As a result, the reaction temperature could be raised moderately to accelerate the reaction rate. For example, Li et al.

* Corresponding authors.
E-mail addresses: nanjin@ytu.edu.cn (N. Jin), yczhao@ytu.edu.cn (Y. Zhao).

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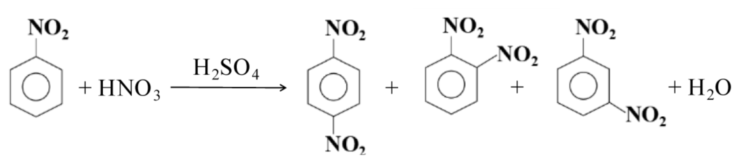
(Li et al., 2017; Li et al., 2022) realized the nitration of 2-ethylhexanol and 3-[2-chloro-4-(trifluoromethyl) phenoxy] benzoic acid at room temperature in continuous flow microreactors. Compared with the commonly used temperature of around 273 K in batch reactors, not only shorter reaction times (seconds or minutes) were acquired in microreactors, but also operating difficulties and energy consumption could be decreased. Beside the above-mentioned merits, reduced amounts of acids and improved conversion as well as yields of target products are often obtained when performing the reaction continuously in a microreactor. Russo et al. (Russo et al., 2019) realized a high conversion of 97% in the nitration of benzaldehyde in a microreactor, achieving the theoretical limit of conversion attainable under kinetic control. Wen et al. (Wen et al., 2017) used a microreactor for the continuous nitration of trifluoromethoxybenzene in which the conversion was up to 99.6% and the selectivity of the main product (4-(trifluoromethoxy)nitrobenzene) reached 90.97% within 2.4 min. Furthermore, microreactors have shown advantages in the characterization of nitration kinetics, mainly due to the high degree of controllability over the temperature and concentrations in the multiphase microenvironment (Cui et al., 2022; Li et al., 2017; Li et al., 2022; Song et al., 2022a; Song et al., 2022b).

Among various nitroaromatic compounds, *m*-dinitrobenzene (*m*-DNB) is an important basic industrial chemical, which mainly used to synthesize *m*-phenylenediamine and *m*-nitroaniline (Duan et al., 2022; Zhao et al., 2007). *m*-DNB is generally prepared by nitration of nitrobenzene (NB), as expressed by Eq. (1). In this process, other two isomers, i.e., *o*-dinitrobenzene (*o*-DNB) and *p*-dinitrobenzene (*p*-DNB), are also produced.



(1)

In the industrial production, nitration of nitrobenzene is generally carried out in tank reactors at 50–90 °C with the reaction time of 2–6 h, the molar ratios of 0.9–2.2:1 and 1.01–1.12:1 for H_2SO_4/HNO_3 and HNO_3/NB, respectively (Gao et al., 2020; Xu et al., 2018; Zhang et al., 2020). Under such conditions, the products, *o*-DNB, *m*-DNB, *p*-DNB, are yielded in a ratio of around 12:85:3. As discussed above, the fundamental drawbacks of macroscale reactors (e.g., low space time yield, long reaction time and high safety risk) still hinder the process improvement towards economically viable and environmentally sustainable manufacturing.

Motivated by the compelling advantages of microreactors, a continuous flow microreactor system was developed to perform the safe and efficient nitration of nitrobenzene in this work. The reaction performance was studied in relation to reaction temperature, sulfuric acid strength, molar ratio of substrates and catalysts, flow rate and the residence time. According to experimental results, a kinetic model was developed, based on which the kinetic parameters, such as the reaction rate constant, pre-exponential factor and activation energy, were obtained. The developed kinetic model was further validated by comparing the predicted data with the experimental results, and proved applicable even though the reaction conditions were changed.

2. Experimental

2.1. Chemicals

Nitrobenzene ($C_6H_5NO_2$, 99%), fuming nitric acid (HNO_3, 98%) and sulfuric acid (H_2SO_4, 98%) were purchased from Sinopharm Chemical Reagent Beijing Co., Ltd. Methanol (CH_3OH, HPLC) was procured from Merck Drugs Biotechnology.

2.2. Experimental setup and procedures

Fig. 1 shows the continuous flow experimental setup for nitrobenzene nitration. It contained two high-pressure syringe pumps (Harvard 11 ELITE, USA, 1.26 pL/min-88.4 mL/min, ±0.5%) equipped with two glass syringes (10 mL), two feeding microchannels (fluorinated ethylene propylene (FEP) capillary with an inner diameter of 0.5 mm), a T-type micromixer (polytetrafluoroethylene (PTFE); inner diameter of 0.5 mm), and a microreactor (FEP capillary with an inner diameter of 0.5 mm). The feeding microchannels, micromixer and microreactor were immersed into a thermostatic water bath (Julabo GmnH CORIO CD-200F, Germany, −20–150 ℃, ±0.1 ℃) with controlled temperature. The outlet of the microreactor was inserted into a sampling tube containing an ice-water mixture to quench the reaction, which was proved effective on account of the significant reduction of temperature and concentrations. After separating the aqueous phase, the collected organic phase was rapidly diluted (mass ratio at 1:2000) in the mobile phase (methanol/water mixture with a mass ratio of 1:1) and then analyzed by liquid chromatography.

To visualize the flow pattern of the nitration system in microreactor, a high-speed CCD video camera (Phantom R311, USA, 24–650000 fps frame rate, 1280 × 800 pixels, 20 μm pixel size) and a stereo microscope (Olympus SZX16, USA, 2.1x-690x magnification, 16.4x zoom) were employed, as shown in Fig. 2. The testing microreactor was inserted in a visualization cell connected to a water bath, in order to maintain the desired temperature inside the microreactor. The details of this system have been described in the previous publication (Ji et al., 2022).

For each run, the system was running for at least 5 min before collecting samples or recording flow patterns, to ensure the establishment of a stable and reliable system. All the experiments were repeated at least twice.

2.3. Analytical procedure

The collected and diluted organic phase was analyzed by HPLC (Waters e2695, USA) equipped with an Agilent Eclipse Plus C18 column (5 μm, 4.6 × 250 mm) and an UV detector. The analysis conditions were as follows: column temperature at 30 ℃, detector wavelength at 254 nm, and mobile phase flow rate at 500 μL/min.

The composition of product was then quantified by the external standard method, and the molar concentration of various substances was determined by the corrected peak area normalizing method. All the samples were analyzed at least twice, and the average relative errors of replication were both less than 3%.

2.4. Some definitions

The reaction characteristics of nitrobenzene nitration in microreactors were evaluated under different operating conditions, i.e., by changing the residence time (τ), reaction temperature (T), sulfuric acid strength (φ), molar ratio of H_2SO_4 to HNO_3 ($M_{\mathrm{ratio},\,1}$), flow rate ratio of

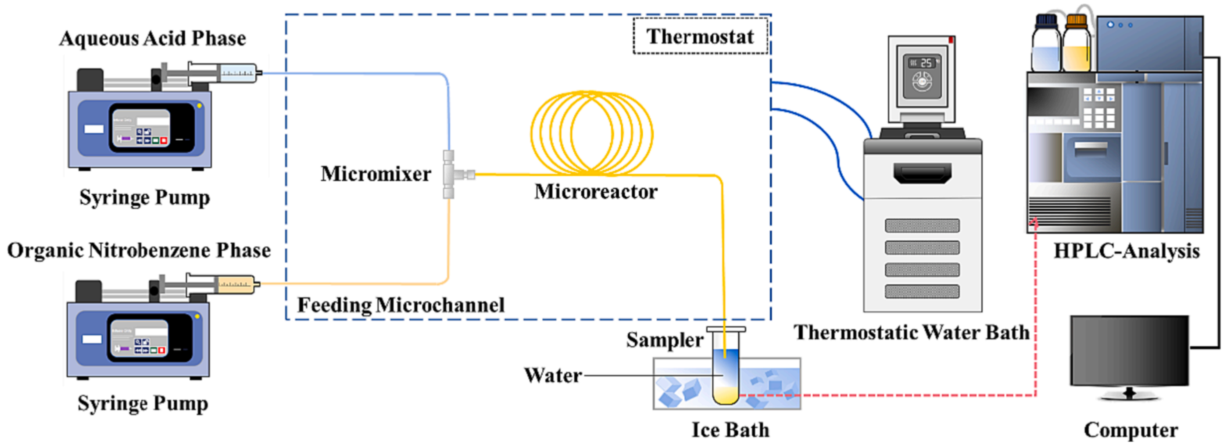


Fig. 1. Schematic of the microreactor system for nitrobenzene nitration.

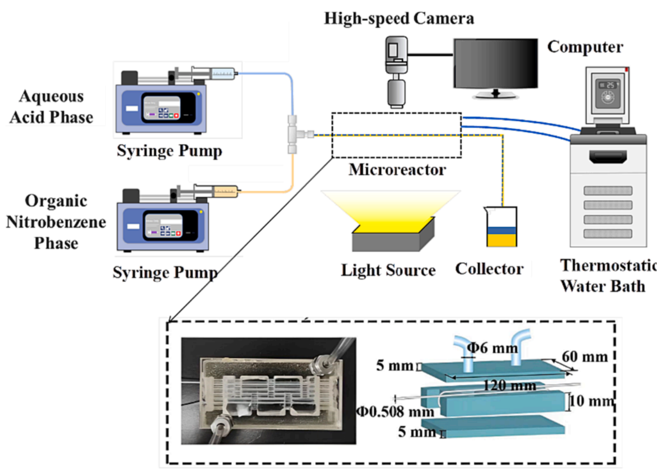


Fig. 2. Schematic of the microreactor system for visualizing and imaging.

aqueous phase to organic phase (Q_{aq}/Q_{org}, equivalent to molar ratio of nitrobenzene to HNO_3, $M_{ratio,\ 2}$), and total volume flow rate (Q_t). They are defined as:

$$\tau = \frac{V_t}{Q_t} \tag{2}$$

$$\varphi = \frac{W_{H_2SO_4}}{W_{H_2SO_4} + W_{H_2O}} \times 100\% \tag{3}$$

$$M_{ratio,1} = \frac{N^0_{H_2SO_4}}{N^0_{HNO_3}} \tag{4}$$

$$M_{ratio,2} = \frac{N^0_{NB}}{N^0_{HNO_3}} \tag{5}$$

where V_r is the volume of microreactor; Q_t, Q_{aq} and Q_{org} are the flow rates of two phases, the aqueous and organic phase, respectively; $W_{H_2SO_4}$ and W_{H_2O} are the mass fractions of H_2SO_4 and water in the mixed acid, respectively; $N^0_{H_2SO_4}$, $N^0_{HNO_3}$ and N^0_{NB} are the initial molar numbers of H_2SO_4, HNO_3 and nitrobenzene before reaction, respectively.

The conversion of nitric acid (X), product selectivity (S) and yield (Y) are defined as:

$$X = \frac{(w_{o-DNB} + w_{m-DNB} + w_{p-DNB})/M_{DNB}}{(w_{o-DNB} + w_{m-DNB} + w_{p-DNB})/168.11 + w_{NB}/M_{NB}/M_{ratio,2}} \times 100\% \tag{6}$$

$$S_i = \frac{w_i}{w_{o-DNB} + w_{m-DNB} + w_{p-DNB}} \times 100\% (i = o\text{ - DNB},\ m\text{ - DNB},\ p\text{ - DNB}) \tag{7}$$

$$Y_i = X \cdot S_i (i = o\text{ - DNB},\ m\text{ - DNB},\ p\text{ - DNB}) \tag{8}$$

where $w_{o\text{-DNB}}$, $w_{m\text{-DNB}}$, $w_{p\text{-DNB}}$, w_{NB} are the mass concentrations of *o*-DNB, *m*-DNB, *p*-DNB, and nitrobenzene in the product, respectively. M_{DNB} and M_{NB} are the molecular weights of dinitrobenzene and nitrobenzene, respectively.

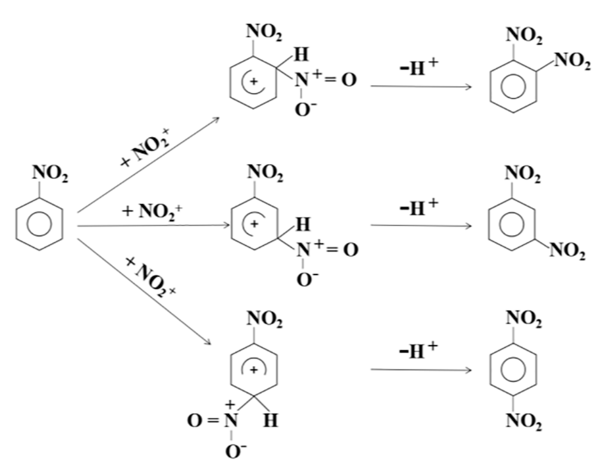
3. Results and discussion

3.1. Fundamentals of heterogeneous nitration of nitrobenzene in microreactor

3.1.1. Reaction regime

On the basis of the results collected during the previous investigations, nitronium ion (NO_2^+) is the real nitrating species (Guo et al., 2023; Li et al., 2022; Rahaman et al., 2010; Song et al., 2022a; Song et al., 2022b), hence a simplified reaction scheme for the nitration of nitrobenzene has been proposed as shown in Eqs. (9)-(10) (Rahaman et al., 2010). Firstly, the reactive species NO_2^+ is released through the protonation of nitric acid in the presence of sulfuric acid. Then, NO_2^+ attacks the aromatic nucleus to generate an intermediate namely σ-complex. The different substitution position of the introduced nitro group would give rise to three isomers. Finally, the σ-complex rapidly deprotonates and thereby produce dinitrobenzene.

$$HNO_3 + H_2SO_4 \rightarrow NO_2^+ + H_2O + HSO_4^- \tag{9}$$



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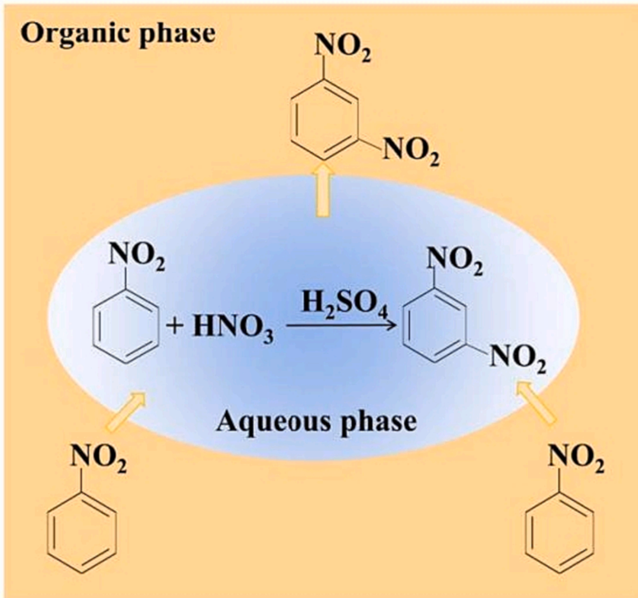


Fig. 3. The heterogeneous nitration process of nitrobenzene.

For the operating conditions used in industries, nitration of nitrobenzene is a typical liquid–liquid heterogeneous process, in which the above reactions are coupled strongly with mass transfer between phases. Considering that the reaction is generally accepted to take place in the aqueous phase (Zaldivar et al., 1995), the heterogeneous nitration process could be divided into three steps, as shown in Fig. 3: (1) the transfer of nitrobenzene from the organic phase to the aqueous phase across the two-phase interface, (2) the reaction between nitrobenzene and nitric acid to form dinitrobenzene under the catalysis of sulfuric acid in the aqueous phase, (3) the transfer of dinitrobenzene from the aqueous phase to the organic phase across the two-phase interface.

Through preliminary experiments of this process in microreactor, it was found that the isolation of the solid product (dinitrobenzene) from the organic phase was problematic. To solve such blocking problems, the reaction was normally performed with temperature higher than melting point of the solid, or with a large amount of inertia solvents (Li et al., 2022; Zhang et al., 2020). This inevitably increases energy consumption and subsequent purification difficulties. In this work, an excess of nitrobenzene was used as a solvent to dissolve the dinitrobenzene produced. This makes it possible to decrease the reaction temperature, thereby reducing the formation of by-products and improving the production safety. Referring to the previous studies (Modak and Juvekar, 1995; Rahaman et al., 2007, 2010; Zaldivar et al., 1995), the reaction temperature was set at 25–65 ℃, higher than which nitric acid began to decompose (Russo et al., 2017). In addition to acting as a solvent, excess nitrobenzene is also beneficial for sufficient conversion of nitric acid and reduced waste-acid discharge accordingly.

To determine a suitable amount of nitrobenzene, the solubility of dinitrobenzene in nitrobenzene needs to be known, as measured using the following procedure. A mixture of *m*-dinitrobenzene and nitrobenzene was stirred for 12 h at 25–65 ℃ and then stood for 12 h. After separating the organic phase with the undissolved solid, the content of *m*-dinitrobenzene in the organic phase was analyzed by HPLC. It was found that the solubility of *m*-dinitrobenzene in nitrobenzene (in molar ratio) was 0.303, 0.752 and 1.510 mol/mol at 25, 45 and 65 ℃., respectively. The reasonable molar ratio of nitrobenzene to nitric acid, correspondingly, is 4.3:1, 2.3:1 and 1.7:1, respectively. However, preliminary experiments showed that on account of a lower temperature at the microreactor outlet than that in the microreactor, the solid products were precipitated after the molar ratio was reduced to 3:1. Therefore, the molar ratio of nitrobenzene to nitric acid was set at 4.3:1 in this work to avoid blocking the microreactor under all conditions.

3.1.2. Flow pattern

A stable and controllable two-phase flow in microreactors is essential to regulate the nitration system. Hydrodynamic experiments were conducted with nitrobenzene and the sulfuric acid solution (at a concentration of 74 wt%, corresponding to the absolute concentration of H_2SO_4 in the actual reaction). The volumetric flow rate ratio of the aqueous phase to organic phase (Q_{aq}/Q_{org}) was varied from 0.1 to 0.5 and the total flow rate (Q_t) was from 150 μL/min to 5500 μL/min. It was found from Fig. 4 that aqueous phase appeared as the dispersed phase and organic phase served as the continuous phase. There were four types of flow pattern that existed under different conditions, namely slug flow, droplet flow, irregular droplet flow and slug-droplet flow. Among them, slug flow and droplet flow are stable and controllable, making them suitable for nitration process.

Fig. 5 shows the flow pattern map of the involved experimental system in the microreactor under different temperatures. As expected, slug flow had the largest operating window with a relatively high Q_{aq}/Q_{org}. When $Q_{aq}/Q_{org} \geq 0.3$ and Q_{aq} less than 400 μL/min, the investigated system almost appeared as slug flow. Decreasing Q_{aq}/Q_{org}, droplet flow began to occur at a medium Q_{aq}. When Q_{aq} was higher than 450 μL/min, the flow pattern became irregular. Temperature had little effect on the flow pattern. Even though the temperature increased, the flow pattern was still controllable. In the following sections, the nitration experiments were conducted and the results presented are in the slug or droplet flow zone.

3.2. Reaction characteristics of nitration in microreactors

3.2.1. Effect of residence time and reaction temperature

The nitration characteristics were firstly studied regarding the influence of two essential parameters, residence time and reaction temperature, with sulfuric acid strength, total volume flow rate, molar ratios of H_2SO_4 to HNO_3 and nitrobenzene to HNO_3 being fixed at 85%, 98.17 μL/min, 4.3 and 2, respectively. As shown in Fig. 6, with increasing the residence time or temperature, the conversion of nitric acid increased. Under all these conditions, the reaction rate was fast at first and then tended to slow down in a short time. More than 40% nitric acid was converted within 30 s at 45 °C, and a higher value with almost 60% was reached at 65 °C. When the reaction time was extended to 600 s under 65 °C, the nitric acid conversion increased to 83.53%.

At the beginning of reaction period, the concentrations of H_2SO_4 and HNO_3 were relatively high to form abundant NO_2^+, which induced the reaction rapidly. As the reaction time was increased, their concentrations decreased and the reaction rate slowed down, then the HNO_3 conversion increased less significantly. The effect of temperature on the conversion of HNO_3 can be explained from the aspects of chemical reaction rate and mass transfer rate. On the basis of Arrhenius equation, the reaction rate increases with increasing temperature. For liquid–liquid non-homogeneous reaction, the mass transfer rate also affects the overall reaction rate (i.e., if not fully under kinetic control). With increasing temperature, the mass transfer rate of aromatic from pure organic phase to the aqueous acid phase also increases (Ravikumar Bandaru and Ghosh, 2011). Therefore, a higher conversion of HNO_3 was obtained at a higher temperature under the same residence time.

Fig. 6(b) and 6(c) show the influence of residence time and temperature on the product distribution. Under the investigated conditions, three products (*m*-DNB, *o*-DNB and *p*-DNB) were detected. Among them, *m*-DNB accounted for the largest proportion with 83–90%, *o*-DNB came second with 8–13%, while that of *p*-DNB was only 2–4%. The product selectivity is significantly impacted by the stability of the σ-complex (Norman and Radda, 1961; Rahaman et al., 2010). When an aromatic ring has an electron-giving group, it is easier for positive charges to be distributed more evenly on the σ-complex, which increases both its stability and the pace of nitration reaction while *o*-nitration or *p*-

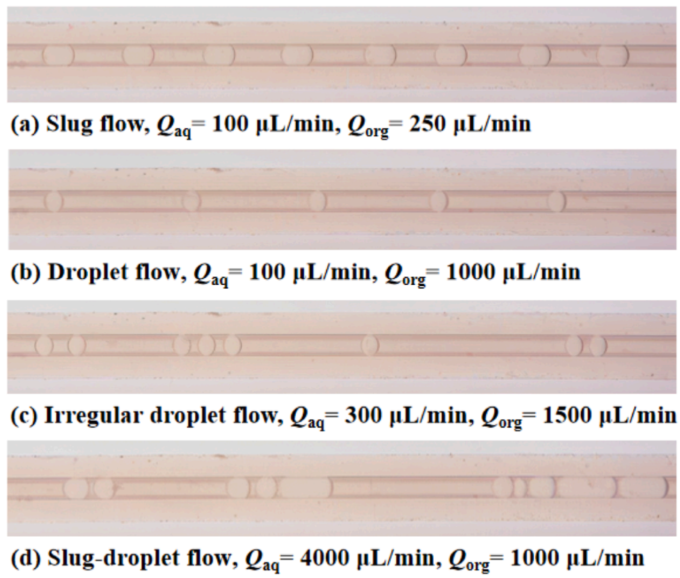


Fig. 4. Typical photographs of the two-phase flow in the microchannel for nitrobenzene-sulfuric acid system.

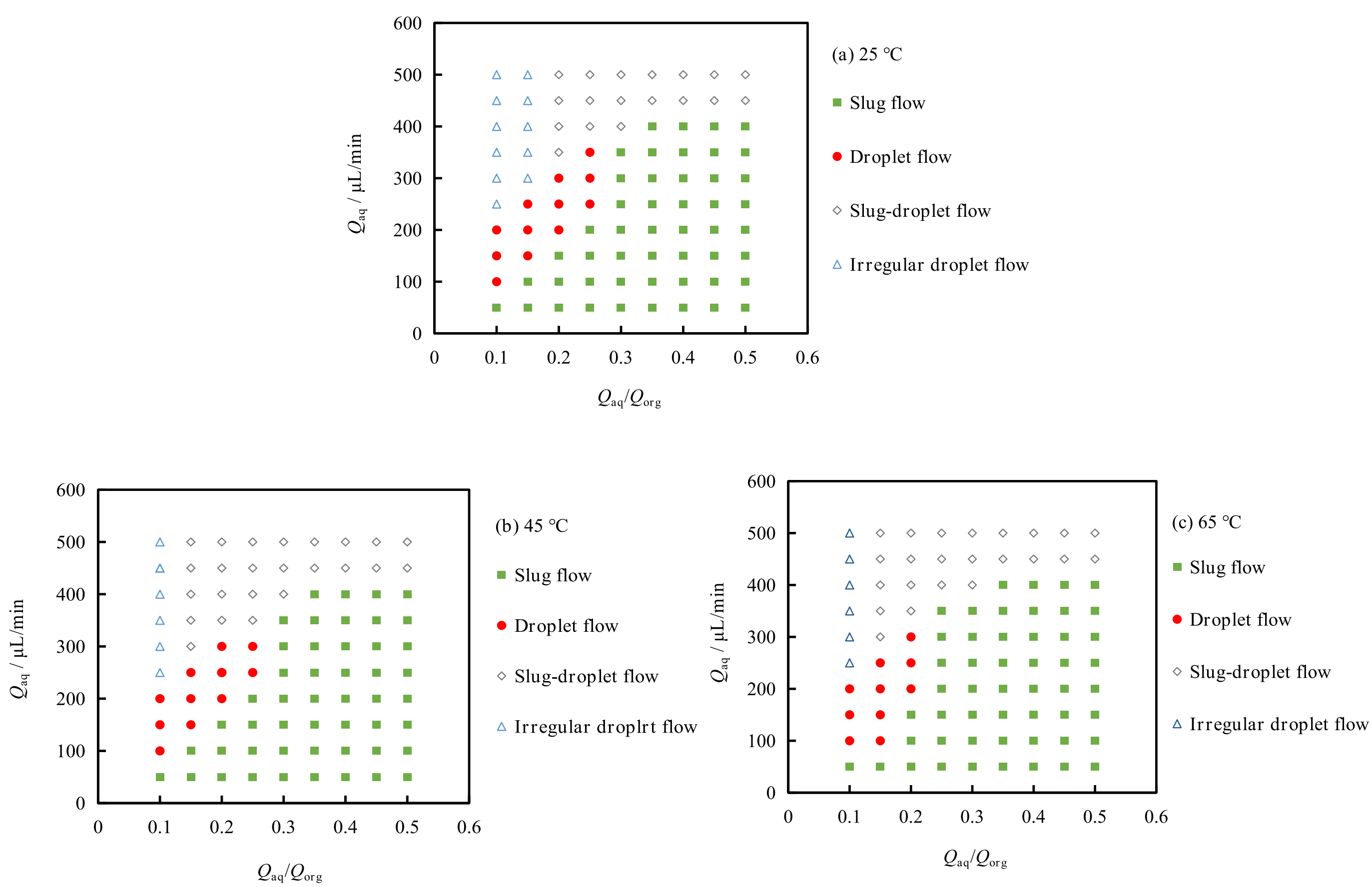


Fig. 5. Effect of two-phase flow rate and temperature on flow patterns for nitrobenzene-sulfuric acid system.

nitration products have a tendency to be formed. However, the stability and nitration rate of the σ-complex diminish when an electron-absorbing group is present on the aromatic ring, thereby increasing the selectivity of the *m*-nitration product. Since nitrobenzene already contains a nitro group, a powerful electron-absorbing group, the main product of nitrobenzene nitration is *m*-dinitrobenzene.

The selectivity of *m*-DNB decreased slightly with temperature rise, while had almost no change with residence time. The reason is that *o*-DNB and *p*-DNB are parallel by-products rather than sequential ones, and prolonging the residence time does not change the relative reaction rates of the three reactions. However, the side reactions to generate ortho and para products have higher activation energies than the main reaction, which results in these reactions exhibiting higher temperature sensitivities.

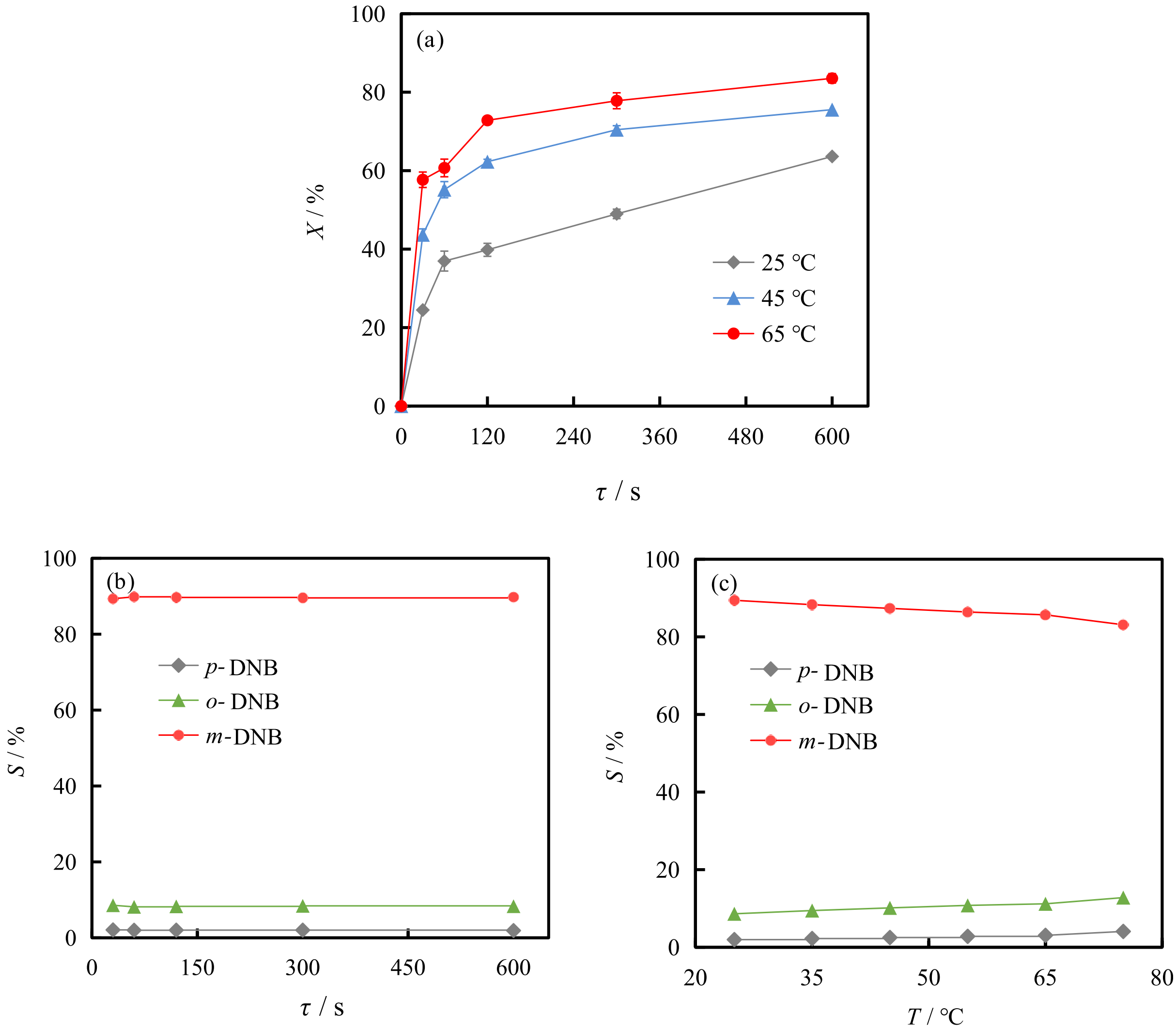


Fig. 6. Effect of residence time and temperature on (a) conversion and (b-c) selectivity in nitration of nitrobenzene. (a)-(c): $\varphi = 95\%$, $Q_t = 98.17$ μL/min, $M_{\text{ratio, 1}} = 2$, $M_{\text{ratio, 2}} = 4.3$; (b) $T = 25$ ℃; (c) $\tau = 600$ s.

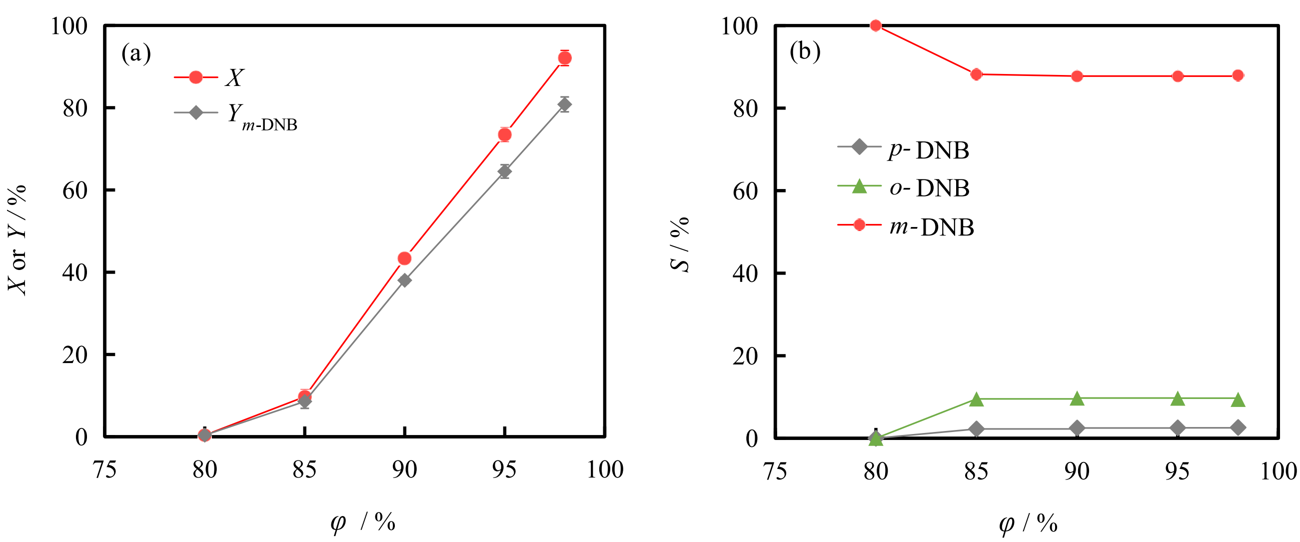


Fig. 7. Effect of sulfuric acid strength on (a) conversion, yield and (b) selectivity in nitration of nitrobenzene. ($\tau = 600$ s, $T = 45$ ℃, $Q_t = 98.17$ μL/min, $M_{\text{ratio, 1}} = 2$, $M_{\text{ratio, 2}} = 4.3$).

3.2.2. Effect of sulfuric acid strength

Varying the sulfuric acid strength (i.e., sulfuric acid content in the aqueous phase), while keeping other parameters fixed ($\tau = 600$ s, $T = 45$ ℃, $Q_t = 98.17$ μL/min, $M_{ratio,\ 1} = 2$, $M_{ratio,\ 2} = 4.3$), the conversion and selectivity varied as depicted in Fig. 7.

With increasing the sulfuric acid strength, the conversion of HNO_3 increased markedly. At 98% H_2SO_4 strength, 92.08% HNO_3 was converted, while nitration scarcely occurred within 10 min when the H_2SO_4 strength decreased to 80%. The great influence of H_2SO_4 strength on the nitration can be explained from the aspects of thermodynamics and kinetics. As discussed above, nitration proceeds through the intermediation of NO_2^+, while its concentration is highly dependent on the strength of sulfuric acid. Actually, except for protonation to generate NO_2^+ (Eq. (11)), there is another equilibrium for nitric acid coexisting simultaneously in aqueous sulfuric acid, i.e., dissociation to NO_3^- (Eq. (12)) (Zaldivar et al., 1995). Spectral analysis has indicated that the equilibrium would switch from formation of NO_3^- to NO_2^+ as the sulfuric acid percent increases (Zaldivar et al., 1995). Besides, sulfuric acid with higher strength can also act as a water absorbent. Its strong ability to absorb water promotes the protonation of HNO_3 and thereby the formation of NO_2^+, as shown in Eq. (9). On the other hand, when H_2SO_4 strength increases, nitrobenzene becomes more soluble in the aqueous phase (Rahaman et al., 2010). Obviously, higher concentrations of both nitronium ion and nitrobenzene can speed up the reaction rate.

$$HNO_3 + H^+ \rightleftharpoons NO_2^+ + H_2O \tag{11}$$

$$HNO_3 \rightleftharpoons NO_3^- + H^+ \tag{12}$$

In the nitration of benzene, the reaction can still proceed at a 70% sulfuric acid strength (Burns and Ramshaw, 2002). Compared with benzene, nitrobenzene carries a nitro group, which is a strong electron-absorbing group that decreases the density of the electron cloud on the benzene ring. The reduced density of electron cloud is not conducive to the occurrence of electrophilic substitution reactions, making the nitration rate of nitrobenzene much lower than benzene. In order for the reaction to proceed as much as possible, the reaction severity needs to be improved. In fact, apart from enhancing the sulfuric acid strength, extending the residence time to 40 min or increasing the temperature to 65 °C at 80% H_2SO_4 strength, more HNO_3 can also be converted (i.e., 8.64% and 5.06% respectively).

The selectivity of *m*-DNB was around 87% for the H_2SO_4 strength of 85–98%, while it was almost 100% at 80% H_2SO_4 strength, maybe due to the fact that there is little reaction and it is difficult to detect the by-products (Fig. 7b).

3.2.3. Effect of molar ratio of H_2SO_4 to HNO_3

Fig. 8 shows the influence of catalyst amount on nitration of nitrobenzene at a given residence time of 600 s, reaction temperature of 45 °C, sulfuric acid strength of 95%, NB/HNO_3 molar ratio of 4.3 and total flow rate of 98.17 μL/min.

An increase in the amount of sulfuric acid leads to an increase in the concentration of NO_2^+ and a consequent increase in the reaction rate. Therefore, the conversion of HNO_3 increased sharply from 18.96% to 93.07% as the H_2SO_4/HNO_3 molar ratio increased from 0.5 to 3.0. Further increasing the molar ratio to 3.5, maybe almost all HNO_3 was converted to NO_2^+ at very high H_2SO_4 strength and thus the conversion of HNO_3 kept around 93%. From Fig. 8(b) it can be seen that the selectivity of *m*-DNB remained 88% under all the investigated molar ratios.

3.2.4. Effect of flow rate ratio of aqueous phase to organic phase

At a given sulfuric acid strength, molar ratio of H_2SO_4 to HNO_3 and total flow rate, the effect of flow rate ratio of aqueous phase to organic phase (Q_{aq}/Q_{org}) is presented in Fig. 9. When Q_{aq}/Q_{org} was decreased from 0.48 to 0.21 (equivalent to $M_{ratio,\ 2}$ of 3 to 7), HNO_3 conversion decreased from 93.02% to 78.07%, and the product selectivity remained almost unchanged.

The alternation of phase ratio directly gives rise to the variations of droplet and slug sizes, and even transition of flow pattern. Just as shown in Fig. 10, decreasing Q_{aq}/Q_{org} at constant Q_t contributed to larger continuous slugs and slightly smaller droplets, which is responsible for the variation of the interfacial area, and accordingly the overall reaction rate. For the mass transfer of nitrobenzene from the pure organic phase to the aqueous phase, it can be assumed that the mass transfer resistance is located on the side of aqueous phase, and the mass transfer rate is associated with the specific interfacial area based on the volume of acid droplet (a_{aq}). As calculated in Table 1, with the decrease of Q_{aq}/Q_{org} from 0.48 to 0.21, a_{aq} increased gradually from 14447 m^2/m^3 to 16623 m^2/m^3. This will improve the mass transfer rate and thereby the apparent reaction rate theoretically. However, the HNO_3 conversion showed an unexpected declination, implying that there ought to exist other contributing factors.

In fact, although it is well recognized that the reaction occurs in the aqueous acid phase, evidences have shown that interfacial reaction also plays a role in determining the rate of heterogeneous nitration (Modak and Juvekar, 1995). The rate of interfacial reaction is proved strongly dependent on the specific interfacial area based on the total volume of two phases (a_t). In the present work, when decreasing Q_{aq}/Q_{org} from 0.48 to 0.21, a_t was reduced from 3952 m^2/m^3 to 2246 m^2/m^3 (Table 1), which slowed down the rate of interfacial reaction, and accordingly the overall rate of nitration. Combing with the above analysis on mass transfer, it can be concluded that the unexpected reduction in conversion of HNO_3 is a consequence of decreased interfacial reaction rate, which counteracts the positive effect of the increased mass transfer rate.

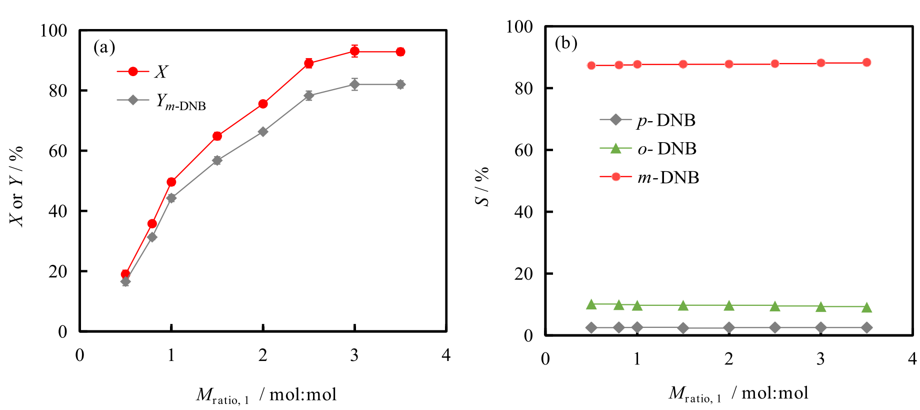


Fig. 8. Effect of molar ratio of sulfuric acid to nitric acid on (a) conversion, yield and (b) selectivity in nitration of nitrobenzene. ($\tau = 600$ s, $T = 45$ ℃, $\varphi = 95\%$, $Q_t = 98.17$ μL/min, $M_{ratio,\ 2} = 4.3$).

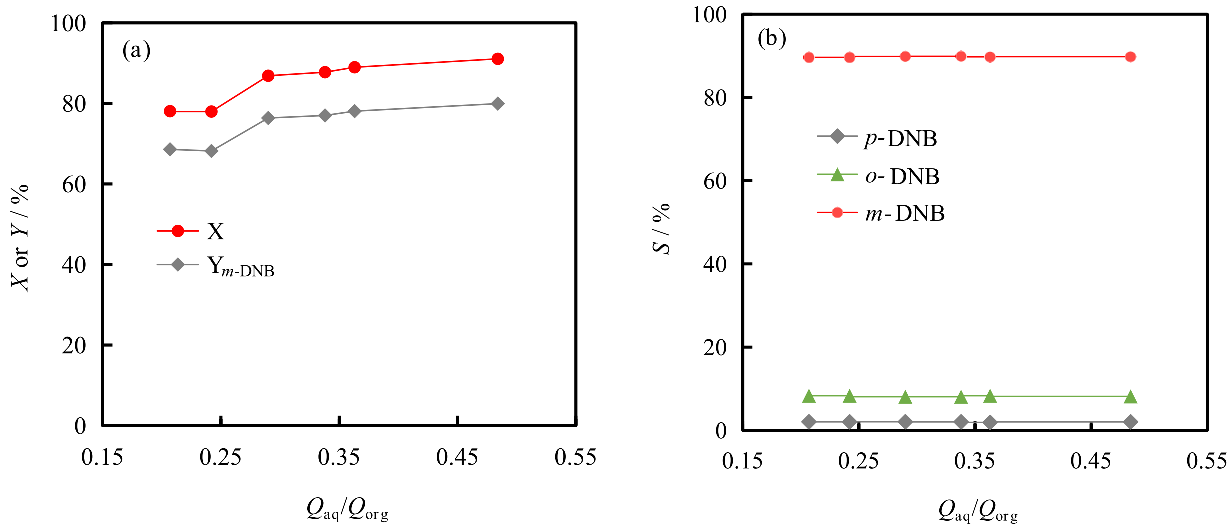
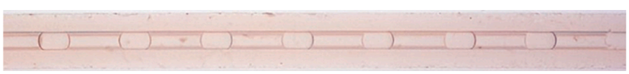
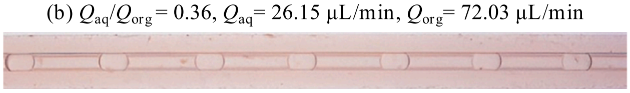


Fig. 9. Effect of flow rate ratio of aqueous phase to organic phase on (a) conversion, yield and (b) selectivity in nitration of nitrobenzene. (τ = 600 s, T = 45 ℃, φ = 98%, Q_t = 98.17 μL/min, $M_{ratio,\ 1}$ = 2).



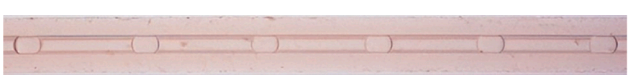
(a) Q_{aq}/Q_{org} = 0.48, Q_{aq} = 32.02 μL/min, Q_{org} = 66.15 μL/min

(b) Q_{aq}/Q_{org} = 0.36, Q_{aq} = 26.15 μL/min, Q_{org} = 72.03 μL/min



(c) Q_{aq}/Q_{org} = 0.34, Q_{aq} = 24.78 μL/min, Q_{org} = 73.39 μL/min

(d) Q_{aq}/Q_{org} = 0.29, Q_{aq} = 22.10 μL/min, Q_{org} = 76.08 μL/min



(e) Q_{aq}/Q_{org} = 0.24, Q_{aq} = 19.13 μL/min, Q_{org} = 79.04 μL/min

(f) Q_{aq}/Q_{org} = 0.21, Q_{aq} = 16.87 μL/min, Q_{org} = 81.31 μL/min

Fig. 10. Micrographs of the two-phase flow under different phase ratios.

Table 1
Specific interfacial area under different Q_{aq}/Q_{org}.

Q_{aq}/Q_{org}	Specific interfacial area based on two phases (a_t)/ m^2/m^3	Specific interfacial area based on acid droplets (a_{aq})/ m^2/m^3
0.48	3951.95	14447.45
0.36	3716.35	15064.64
0.34	3254.10	15256.89
0.29	2910.69	15632.13
0.24	2573.09	15821.67
0.21	2245.74	16622.55

3.2.5. Effect of total volumetric flow rate

The total volume flow rate was altered by adjusting the microreactor length while maintaining the residence time at 60 s. The results (Fig. 11) indicate that the conversion of HNO_3 increased with increased total volume flow rate within limits of 981.75 μL/min, beyond which the HNO_3 conversion remained almost constant around 63%. The total volume flow rate had nearly no impact on the product selectivity.

Besides the phase ratio, the total flow rate can also lead to the variation of interfacial area caused by the droplet size. As shown in Fig. 12, the higher the total flow rate, the smaller the droplet size. At the same time, the two-phase flow pattern turned from slug flow to droplet flow. Table 2 summarizes the interfacial areas based on the volumes of two-phase and the aqueous phase, respectively. With the increase of Q_t from 98.17 μL/min to 981.75 μL/min, a_t and a_{aq} were improved by a factor of 1.31 or 1.16, respectively. Both the increase of a_t and a_{aq} are favorable for speeding up the reaction. In addition, since there was an obvious reduction in droplet size, the enhanced internal circulation within the droplet can also promote the mass transfer performance, which has been demonstrated by many researchers (Tsaoulidis and Angeli, 2016; Yao et al., 2018). When the total flow rate exceeded 981.75 μL/min, these effects became insignificant on the overall conversion which therefor leveled off.

Summarizing the effects of operating conditions on the nitration, an optimized reaction conditions were obtained in microreactors, recorded with temperature of 65 ℃, residence time of 10 min, sulfuric acid strength of 98%, molar ratio of nitrobenzene to nitric acid of 4.3, molar ratio of sulfuric acid to nitric acid of 2, and total flow rate of higher than 981.75 μL/min. Under these conditions, more than 90% nitric acid was converted and 80 % *m*-dinitrobenzene was yielded within 10 min. As discussed above, submillimeter characteristic dimension of microreactor brings not only short diffusion distance (submillimeter scale), but also large interfacial area (higher than 10^3 m^2/m^3) that is particularly important for a heterogeneous process. With enhanced mass transfer performance, microreactor shortens the residence time drastically from the order of hours to minutes, which means a higher space time yield and thereby an improved production efficiency. At the same time, as the solvent of dinitrobenzene, excess nitrobenzene avoids the channel blockage caused by precipitation of solid products. This makes it possible to reduce the reaction temperature below the melting points of solid products, consequently, the energy consumption could be reduced. More importantly, the hazardous by-products (such as nitrophenol, 3-nitrobenzenesulfonic acid, and trinitrobenzene), which often occurred under higher temperatures, were not detected under all conditions. This indicates that the temperature control of microreactor is effective based on its excellent capability of eliminating local “hot spots”, which could help to improve the production safety. Nevertheless, from the aspect of large-scale application, fundamental research on the microreactor scale-up, such as how to well handle the fluid distribution across numerous microchannels in parallel and a multitude of microreactor modules,

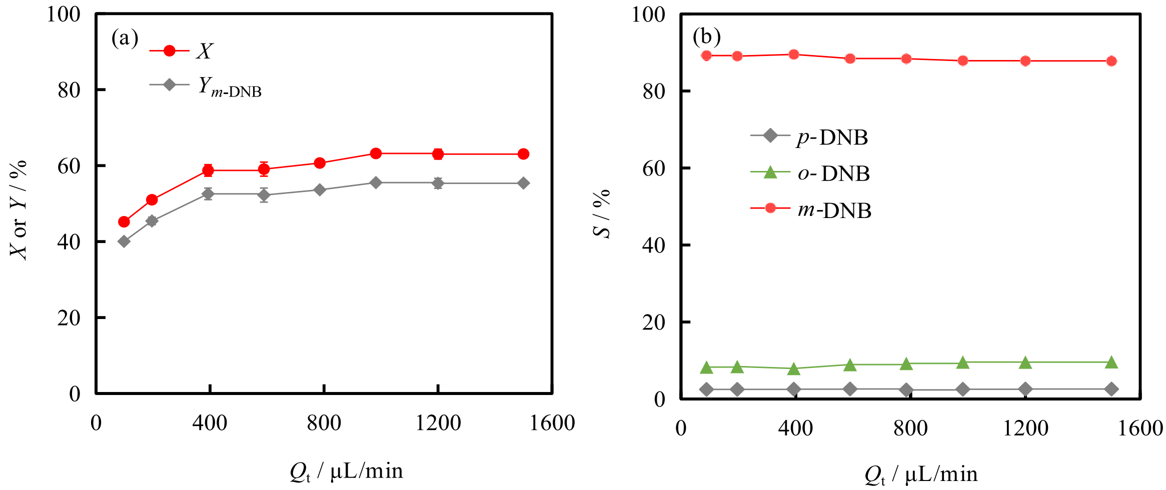


Fig. 11. Effect of the total volumetric flow rate on (a) conversion, yield and (b) selectivity in nitration of nitrobenzene. (τ = 60 s, T = 45 ℃, φ = 95%, $M_{\text{ratio, 1}}$ = 2, $M_{\text{ratio, 2}}$ = 4.3).

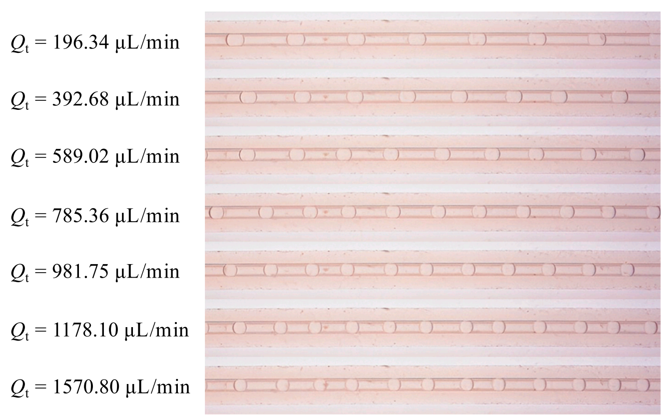


Fig. 12. Micrographs of the flow patterns under different volumetric flow rates at a fixed Q_{aq}/Q_{org} of 0.34.

Table 2
Specific interfacial area under different flow rates.

Q_t / μL/min	Specific interfacial area based on two phases (a_t) / m^2/m^3	Specific interfacial area based on acid droplets (a_{aq}) / m^2/m^3
98.17	4025.75	12761.90
196.34	4147.73	13185.19
392.68	4313.89	13638.34
589.02	4518.83	14086.96
785.36	4764.83	14588.24
981.75	5275.86	14746.99
1178.1	5172.41	15000.00
1570.8	5125.42	14896.55

remains to be further investigated.

3.3. Reaction kinetics of nitration in microreactors

Kinetic data are necessary for reaction parameter optimization in the industrial production. For non-homogeneous nitration, mass transfer and reaction are difficult to decouple entirely, even though a better mixing performance can be obtained easily in microchannels. Nonetheless, by eliminating the effect of mass transfer on kinetic measurement as much as possible, a pseudo-homogeneous model is suitable for establishing the nitration kinetics (Li et al., 2022). In section 3.2.5, we have investigated the effect of total volumetric flow rate on this reaction, according to which the total volume flow rate was set at 1570 μL/min to conduct the kinetic experiments. In addition, lower temperature (15–35 ℃) and shorter residence time (2–5 s) were adopted to slow down the reaction rate, in order to acquire more precise kinetic data. The detailed data are presented in the supplementary information (Figure S1). Under the investigated conditions, the conversion of HNO_3 was no more than 30%, which will help to improve the accuracy of kinetic modelling.

3.3.1. Determination of the observed reaction rate constant based on HNO_3

According to extensive investigations, aromatic nitration generally conforms to second-order kinetics, being first order in aromatics and nitric acid, respectively (Li et al., 2017; Li et al., 2022; Song et al., 2022a; Song et al., 2022b). Therefore, the reaction rate of nitrobenzene nitration can be described as:

$$r = k_{HNO_3} \cdot C_{NB} \cdot C_{HNO_3} \tag{13}$$

where k_{HNO_3} is the observed reaction rate constant based on nitric acid, C_{NB} and C_{HNO_3} are the concentrations of nitrobenzene and total nitric acid in the pseudo-homogeneous phase, respectively.

Integrating Eq. (13) with the initial condition of $X = 0$ at $\tau = 0$, the following equation is finally obtained (not detailed here, see Section S2 in the supplementary information):

$$\frac{\ln\left[\frac{M_{\text{ratio},2} - X}{(1-X)\cdot M_{\text{ratio},2}}\right]}{C^0_{NB} - C^0_{HNO_3}} = k_{HNO_3} \cdot \tau \tag{14}$$

where $C^0_{HNO_3}$ and C^0_{NB} are the initial concentrations of nitric acid and nitrobenzene in the pseudo-homogeneous phase, respectively.

Therefore, a plot of $\frac{\ln\left[\frac{M_{\text{ratio, 2}} - X}{(1-X)\cdot M_{\text{ratio, 2}}}\right]}{C^0_{NB} - C^0_{HNO_3}}$ versus τ is a straight line, the slope of which equals to k_{HNO_3}. As shown in Figure S2, the corresponding plots gave good linear relationships under all sulfuric acid strengths. This verifies that the previously assumed second-order kinetics is reasonable.

Based on the slopes of the straight lines in Figure S2, the relationship between k_{HNO_3} and temperature at different sulfuric acid strengths was obtained, as shown in Fig. 13. k_{HNO_3} for nitration of nitrobenzene was in the range of 4 × 10^{-4} L/(mol·s) to 1.3 × 10^{-2} L/(mol·s). This value is much lower than that for nitration of nitrotoluene (0.35–1.14 L/(mol·s)), and has a slight difference from that for nitration of chlorobenzene (4.39 × 10^{-4}-2.1 × 10^{-2} L/(mol·s)). It reflects the nitration difficulties between different aromatics. It can be also seen that k_{HNO_3} was dependent on not only temperature, but also sulfuric acid strength. This is due to the nitration reaction of aromatics is essentially the reaction of aromatics with NO_2^+, the concentration of which is strongly

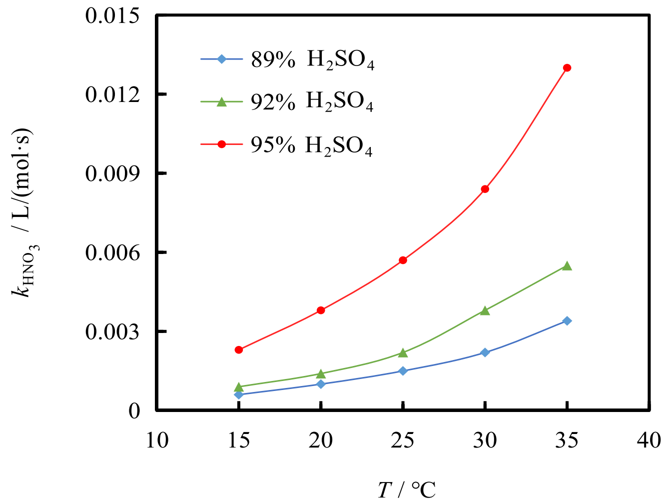


Fig. 13. Variation of the experimental values of k_{HNO_3} with different sulfuric acid strengths at different temperature levels.

related to the sulfuric acid strength (Zaldivar et al., 1995). Therefore, it is necessary to study the reaction rate constant based on NO_2^+.

3.3.2. Determination of the real reaction rate constant based on NO_2^+

When the reaction rate is expressed by using the concentration of NO_2^+, Eq. (13) can be transformed into Eq. (15):

$$r = k_2^* \cdot C_{NB} \cdot C_{NO_2^+} \tag{15}$$

where k_2^* is the observed rate constant based on NO_2^+, and $C_{NO_2^+}$ is the NO_2^+ concentration in the pseudo-homogeneous phase. In the mixed acid solution, the effective concentration of each substance differs from the calculated concentration, requiring the introduction of activity coefficient to correct the concentration of each substance in the system. Based on the transition-state theory, the reaction rate is expressed as follows:

$$r = k^* \cdot C_{NB} \cdot C_{NO_2^+} \cdot \frac{\gamma_{NB} \cdot \gamma_{NO_2^+}}{\gamma^*} \tag{16}$$

where k^* is the real rate constant expressed for NO_2^+, and γ_{NB}, $\gamma_{NO_2^+}$, and γ^* are the activity coefficients for nitrobenzene, NO_2^+, and the transition-state intermediate, respectively.

On account of the difficulties to obtain the activity coefficient, Marziano et al. (Marziano and Sampoli, 1983) utilized an activity coefficient function (namely M_c) to describe the relationship between the activity coefficient with sulfuric acid concentration (Marziano et al., 1998), based on which the nitration rate equation can be expressed as follows (not detailed here, see Section S3 in the supplementary information):

$$r = k^* \cdot C_{NB} \cdot C_{NO_2^+} \cdot 10^{-n_{NB} \cdot M_c} \tag{17}$$

where n_{NB} is a thermodynamic parameter.

To obtain $C_{NO_2^+}$, the equilibrium constants of Eq. (11) and Eq. (12), denoted as $K_{NO_2^+}$ and K_{HNO_3}, are introduced, then it is finally obtained that (not detailed here, see Section S3 in the supplementary information):

$$\lg Z = \lg k^* - n_{NB} \cdot M_c \tag{18}$$

$$Z = k_{HNO_3} \cdot \left[\frac{K_{HNO_3}}{C_{H^+} \cdot 10^{-0.571 \cdot M_c}} + \frac{C_{H^+} \cdot 10^{-2.542 \cdot M_c}}{C_{H_2O} \cdot K_{NO_2^+}} + 1 \right] \cdot \frac{C_{H_2O} \cdot K_{NO_2^+}}{C_{H^+} \cdot 10^{-2.542 \cdot M_c}} \tag{19}$$

where C_{H^+} and C_{H_2O} are the concentrations of H^+ and H_2O in the pseudo-homogeneous phase, respectively.

Table 3
Values of n_{NB} and k^* at different temperatures.

T / K	n_{NB}	k^* / L/(mol·s)
288	0.756	2.917×10^{-10}
293	0.764	4.975×10^{-10}
298	0.770	7.797×10^{-10}
303	0.771	1.396×10^{-9}
308	0.784	1.944×10^{-9}

Plotting lg$Z$ with -M_c as the horizontal coordinate gives a straight line. The resulted fitted line has an intercept of lgk^* and a slope of n_{NB}. As shown in Figure S3, the data obtained were well linear, with parallel straight lines at different temperatures. The values of n_{NB} and lgk^* obtained at different temperatures are presented in Table 3. k^* was in the range of 2.9×10^{-10}-1.9×10^{-9} L/(mol·s), much lower than k_{HNO_3} by 6–7 orders of magnitude. The value of n_{NB} remained essentially unchanged, which is consistent with other studies (Li et al., 2022). k^* increased with increasing temperature, while it was independent of sulfuric acid strength. This means that the use of k^* can decouple the sulfuric acid strength from the reaction rate constant, hence is commonly regarded as the intrinsic reaction rate constant.

Based on the value of k^* at different temperatures, the activation energy for the reaction can be calculated using the Arrhenius equation:

$$\ln k^* = -\frac{E_a}{R \cdot T} + \ln A \tag{20}$$

where E_a is the activation energy for the reaction of nitrobenzene nitration, R is the molar gas constant, and A is the pre-exponential factor.

By plotting $1/T$ versus lnk^* (Fig. 14), the resulted fitted line has an intercept of ln$A$ and a slope of $-\frac{E_a}{R}$. The calculated activation energy and pre-exponential factor were 71.229 kJ/mol and 2456.27 L/(mol·s), respectively.

According to literature data, the activation energies reported for the nitration of several compounds are shown in Table 4. The existence of nitro functional group, an electron-absorbing group, makes the nitration of nitrobenzene and nitrotoluene more difficult than benzene and other aromatics with electron-giving groups. Therefore, the corresponding activation energies are relatively high.

Based on the estimated Ea and A, k_{HNO_3} and then the conversion of nitric acid can be calculated from Eq. (18) and Eq. (14). We compared the experimental and predicted values and the results are shown in Fig. 15. The predicted values were found to be in good agreement with the experimental results, with the relative error less than 10% and 15% for k_{HNO_3} and the nitric acid conversion, respectively.

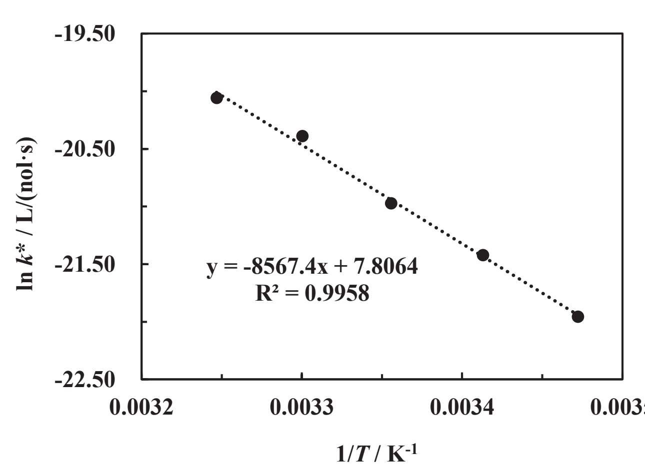


Fig. 14. Arrhenius plot of lnk^* versus $1/T$.

Table 4
E_a for the nitration of different compounds in the literature.

Compound	Ea / kJ/mol
Mesitylene	30.558 (Marziano et al., 1998)
Toluene	35.162 (Marziano et al., 1998)
Benzene	38.511 (Marziano et al., 1998)
Chlorobenzene	46.460 (Marziano et al., 1998)
p-Nitrotoluene	50.207 (Song et al., 2022b)
Nitrobenzene	71.229 (this work)
o-Nitrotoluene	72.358 (Song et al., 2022a)

3.3.3. Validation of the pseudo-homogeneous phase assumption

To validate the assumption of pseudo-homogeneous phase, the Hatta number (*Ha*), the ratio of the reaction rate to the mass transfer rate, was calculated. If *Ha* less than 0.3, the reaction is kinetically controlled and thereby the pseudo-homogeneous hypothesis holds (Li et al., 2022).

$$Ha = \frac{\sqrt{k_{HNO_3} \cdot D_{NB} \cdot C_{HNO_3}}}{k_L} \quad (21)$$

where D_{NB} is the diffusion coefficient of nitrobenzene in the aqueous phase,k_L is the overall mass transfer coefficient of nitrobenzene from the organic phase to the aqueous phase.

The estimated results are shown in Table 5. *Ha* was less than 0.3 under all the investigated conditions, which indicates that the mass transfer rate is much greater than the reaction rate. Since the reaction proceeds slowly, it is kinetically controlled and the assumption of pseudo-homogeneous phase holds. *Ha* increased with increasing temperature and sulfuric acid strength. But since the mass transfer rate is already very fast in this system, increasing the temperature and sulfuric acid strength has a greater effect on the reaction rate than on the mass transfer rate. This is consistent with previous research findings (Li et al., 2022; Modak and Juvekar, 1995).

3.3.4. Validation of the kinetic model

To further verify the applicability of the kinetic model, additional experiments were carried out at 90.5 wt% and 93.5 wt% sulfuric acid strength, with reaction temperatures of 20 °C and 30 °C, respectively. The variation of HNO_3 conversion along with residence time is shown in Fig. 16. The predicted values did not differ significantly from experimental ones. It demonstrates that the kinetic model is applicable even though the sulfuric acid strength is changed. The model will help to gain insights into the whole process of nitrobenzene nitration and will be used to optimize the operating conditions.

Table 5
Diffusion and mass transfer coefficients of nitrobenzene and the corresponding *Ha* values.

T / °C	φ / %	$D_{NB} \times 10^{10}$ / m²/s [a]	$k_L \times 10^5$ / m/s [a]	*Ha*
15	89	3.453	1.236	0.047
	92	3.228	1.156	0.060
	95	3.061	1.088	0.100
20	89	4.053	1.451	0.056
	92	3.830	1.371	0.069
	95	3.639	1.303	0.117
25	89	4.725	1.692	0.064
	92	4.510	1.615	0.080
	95	4.321	1.547	0.132
30	89	5.474	1.960	0.072
	92	5.272	1.887	0.097
	95	5.090	1.822	0.148
35	89	6.302	2.256	0.083
	92	6.121	2.191	0.108
	95	5.954	2.131	0.170

[a] The detailed calculation procedures were offered in the supplementary information (Section S4).

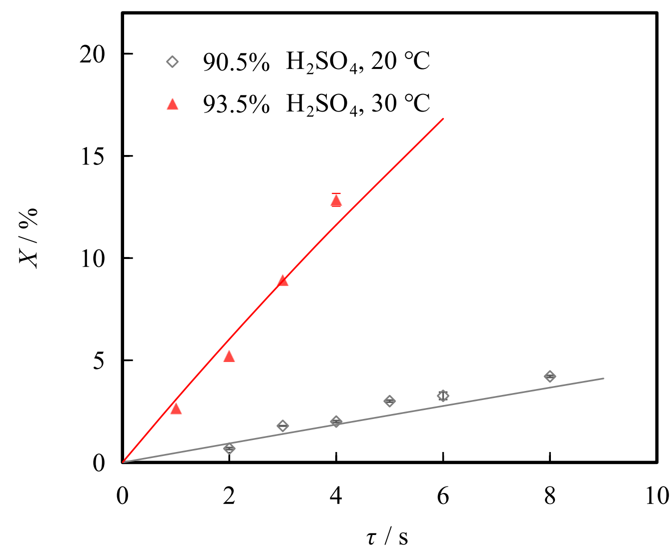


Fig. 16. Comparison between the experimental (symbols) and predicted (lines) values for conversion of HNO_3.

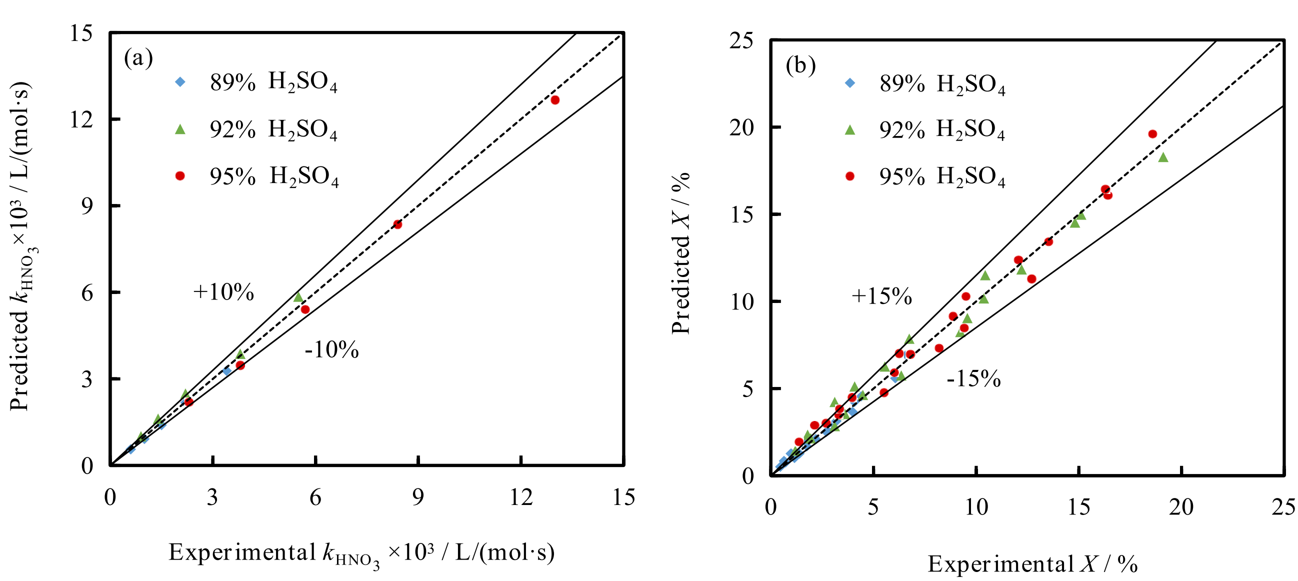


Fig. 15. Comparison of calculated and experimental results for k_{HNO_3} and HNO_3 conversion.

4. Conclusions

To develop a safe and efficient process for the synthesis of *m*-dinitrobenzene, nitration of nitrobenzene with mixed acid was investigated in a continuous flow microreactor. Various operating parameters were firstly examined to quantify their effects on the reaction characteristics. Furthermore, a pseudo-homogeneous kinetic model was developed, based on which the kinetic parameters of the reaction were determined. Finally, the model was validated by comparison with the experimental data.

The conversion of nitric acid increased greatly with increasing reaction temperature, residence time, strength and amount of sulfuric acid. The product selectivity was not sensitive to these factors expect for the temperature. More than 90% nitric acid was converted and 80 % *m*-dinitrobenzene was yielded within 10 min, much shorter than that in the industrial production, which exhibits advantages of the microreactor for process efficiency improvement.

The flow rates of nitrobenzene and mixed acid also influenced the reaction, ascribed to the variation of the interfacial area. Since the measured reaction rate decreased with increasing the phase ratio of nitrobenzene to mixed acid, the reaction was verified to take place not only in the bulk aqueous phase but also at the phase interface. When the total flow rate exceeded 981.75 μL/min, the effect of mass transfer and interfacial reaction can be neglected.

Kinetic experiments were then carried out at a total volume flow rate of 1500 μL/min. The pseudo-homogeneous phase model was used to describe the reaction rate of heterogeneous nitration in the microreactor. By decoupling the sulfuric acid strength from the observed reaction rate constant based on nitric acid, the real reaction rate constant based on NO_2^+, the actual nitrating species, was calculated. Accordingly, the pre-exponential factor and activation energy were obtained as 2456.27 L/(mol·s) and 71.229 kJ/mol, respectively. The developed kinetic model well describes the experimental data. It can provide an important theoretical support for the development of new nitration technology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ces.2023.119198.

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